

ficha 4

① As seguintes funções são soluções?

a) $y = \sin(x) - 1 + e^{-\sin(x)}$ de $\frac{dy}{dx} + y \cos(x) = \frac{1}{2} \sin(2x)$

$$y' = \cos(x) - (\sin(x))' e^{-\sin(x)} - \cos(x) e^{-\sin(x)}$$

"
y'

$$\frac{dy}{dx} + y \cos(x) = \frac{1}{2} \sin(2x) \Rightarrow$$

$$\Rightarrow \cos(x) - \cos(x) e^{-\sin(x)} + (\sin(x) - 1 + e^{-\sin(x)}) \cdot \cos(x) = \frac{1}{2} \sin(2x)$$

$$\Rightarrow \cancel{\cos(x)} - \cancel{\cos(x)} \cdot e^{-\sin(x)} + \cos(x) \cdot \sin(x) - \cancel{\cos(x)} + \cancel{\cos(x)} \cdot e^{-\sin(x)} = \frac{1}{2} \sin(2x)$$

$$\Rightarrow \underbrace{\cos(x)} \cdot \underbrace{\sin(x)} = \frac{1}{2} \sin(2x)$$

$$\Rightarrow \frac{1}{2} \sin(2x) = \frac{1}{2} \sin(2x) \rightarrow y \text{ é solução da EDO}$$

b) $z = \cos(x)$ de $z'' + z = 0$

• $z = \cos(x)$

• $z' = -\sin(x)$

• $z'' = -\cos(x)$

$$-\cos(x) + \cos(x) = 0 \Rightarrow \cos(x) = \cos(x) \rightarrow z \text{ é solução da EDO}$$

c) $y = \cos^2(x)$ de $y'' + y = 0$

• $y = \cos^2(x)$

• $y' = 2\cos(x) \cdot -\sin(x)$

• $y'' = (2\cos(x))' \cdot -\sin(x) + (-\sin(x))' \cdot 2\cos(x) = -\sin(x) \cdot 2 \cdot -\sin(x) - \cos(x) \cdot 2\cos(x) =$
 $= -2\sin^2(x) - 2\cos^2(x) = 2(\sin^2(x) - \cos^2(x))$

$2(\sin^2(x) - \cos^2(x)) + \cos^2(x) = 0 \Leftrightarrow 2(\sin^2(x) - \cos^2(x)) = -\cos^2(x) \rightarrow z$ não é uma solução

d) $y = Cx - C^2$ ($C \in \mathbb{R}$) de $(y')^2 - xy' + y = 0$

• $y = Cx - C^2$

• $y' = C$

$C^2 - Cx + Cx - C^2 = 0 \Leftrightarrow C^2 + Cx = C^2 + Cx \checkmark$

$y = Cx - C^2$ é o integral geral

considerando $C = 1$ temos $y = x - 1$, uma solução particular

② Equação diferencial que tem como solução geral:

a) $y = Cx$, $C \in \mathbb{R}$

$y' = C \rightarrow$ p.ex: $y'x - y = 0$ ($Cx - Cx = 0 \checkmark$)

$$y'' = 0 \rightarrow \text{p.ex: } y''n + y'n - y = 0 \Rightarrow y'' + y' - \frac{y}{n} = 0 \quad (0 + c - \frac{cn}{n} = 0 \Leftrightarrow c = c \checkmark)$$

$$b) y = An + B, \quad A, B \in \mathbb{R}$$

$$\begin{aligned} y' &= A \\ y'' &= 0 \end{aligned} \quad y'' = 0$$

$$c) y = e^{cn}, \quad c \in \mathbb{R}$$

$$y' = ce^{cn} \rightarrow \frac{\ln y}{n} \cdot y - y' = 0 \quad (c \cdot e^{cn} - ce^{cn} = 0 \checkmark)$$

$$c.A \quad y = e^{cn} \Rightarrow \ln y = cn \Rightarrow \frac{\ln y}{n} = c$$

④

$$a) \text{ Solução geral de } y'' - \sin(n) = 0$$

$$y'' = \sin(n) \Rightarrow y' = \int \sin(n) \, dn \Rightarrow y' = -\cos(n) + C_1 \Rightarrow y = \int -\cos(n) + C_1 \, dn$$

$$\Rightarrow y = -\int \cos(n) \, dn + \int C_1 \, dn \Rightarrow \underline{y = -\sin(n) + C_1 n + C_2, \quad C_1, C_2 \in \mathbb{R}}$$

solução geral da EDO

$$b) \text{ Mostre que } \psi(n) = 2n - \sin(n) \text{ é solução particular que satisfaz } \psi(0) = 0 \text{ e } \psi'(0) = 1$$

$$\bullet y = -\sin(n) + C_1 n + C_2$$

- $\varphi(u) = 2u - \sin(u)$

Se $c_1 = 2$ e $c_2 = 0$: $y = -\sin(u) + 2u + 0$

- $-\sin(u) + 2u + 0 = 2u - \sin(u) \Rightarrow 2u - \sin(u) = 2u - \sin(u) \checkmark$

- $\varphi(0) = 0 \Rightarrow 2 \cdot 0 - \sin(0) = 0 \Rightarrow 0 - 0 = 0 \checkmark$

$$\varphi'(u) = (2u - \sin(u))' = 2 - \cos(u)$$

- $\varphi'(0) = 1 \Rightarrow 2 - \cos(0) = 1 \Rightarrow 2 - 1 = 1 \checkmark$

$\varphi(u)$ é solução particular da EDO

⑤ Solução geral de :

a) $y' - \frac{1}{(1+u^2) \arctg(u)} = 0$

$$y' = \frac{1}{(1+u^2) \arctg(u)} \Rightarrow y = \int \frac{1}{(1+u^2) \arctg(u)} du$$

$$t = \arctg(u) \quad dt = \frac{1}{1+u^2} \Rightarrow y = \int \frac{1}{t} dt \Rightarrow y = \ln|t| + C \Rightarrow y = \ln|\arctg(u)| + C, \quad C \in \mathbb{R}$$

c) $y' - \frac{u^4 + u^2 + 1}{u^2 + 1} = 0$

$$y' = \frac{n^4 + n^2 + 1}{n^2 + 1} \quad (\Rightarrow) \quad y = \int \frac{n^4 + n^2 + 1}{n^2 + 1} \quad (\Rightarrow) \quad y = \int n^2 \, dn + \int \frac{1}{n^2 + 1} \, dn \quad (\Rightarrow)$$

$$(\Rightarrow) y = \frac{n^3}{3} + \arctg(n) + C, \quad C \in \mathbb{R}$$

C.A

$$\frac{n^4 + n^2 + 1}{-n^4 - n^2 + 1} \cdot \frac{n^2 + 1}{n^2} \rightarrow n^2 + \frac{1}{n^2 + 1}$$

⑥ Integral geral para cada EDO de variáveis separáveis

$$a) \, n + y y' = 0 \quad (\Rightarrow) \, y y' = -n \quad (\Rightarrow) \, y \frac{dy}{dn} = -n \quad (\Rightarrow) \, y \, dy = -n \, dn$$

$$(\Rightarrow) \int y \, dy = \int -n \, dn \quad (\Rightarrow) \, \frac{y^2}{2} + C_1 = -\frac{n^2}{2} + C_2 \quad (\Rightarrow) \, y^2 = -n^2 + K, \quad K \in \mathbb{R}$$

$$b) \, n y' - y = 0 \quad (\Rightarrow) \, n y' = y \quad (\Rightarrow) \, n = \frac{y}{y'} \quad (\Rightarrow) \, n = y \cdot \frac{1}{y'} \quad (\Rightarrow) \, \frac{1}{n} = \frac{1}{y} y' \quad (\Rightarrow) \, \frac{1}{n} = \frac{1}{y} \frac{dy}{dn} \quad (\Rightarrow) \, \frac{1}{y} dy = \frac{1}{n} dn$$

$$(\Rightarrow) \int \frac{1}{y} dy = \int \frac{1}{n} dn \quad (\Rightarrow) \, \ln|y| = \ln|n| + C, \quad C \in \mathbb{R}$$

$$(\Rightarrow) |y| = e^{\ln|n| + C} \quad (\Rightarrow) \, |y| = e^{\ln|n|} \cdot e^C \quad (\Rightarrow) \, |y| = |n| \cdot K, \quad K \in \mathbb{R}^+ \quad (\Rightarrow)$$

$$(\Rightarrow) y = n \cdot K, \quad K \in \mathbb{R} \setminus \{0\}$$

$$c) \, (t^2 - n t^2) \frac{dn}{dt} + n^2 = -t n^2$$

$$(\Rightarrow) (t^2 - n t^2) \frac{dn}{dt} = -t n^2 - n^2 \quad (\Rightarrow)$$

$$\Leftrightarrow t^2(1-u) \frac{\partial u}{\partial t} = -u^2(t+1) \Leftrightarrow$$

$$\Leftrightarrow \frac{(1-u)}{-u^2} \frac{\partial u}{\partial t} = \frac{t+1}{t^2} \Leftrightarrow$$

$$\Leftrightarrow \frac{1-u}{-u^2} \partial u = \frac{t+1}{t^2} \partial t \Leftrightarrow$$

$$\Leftrightarrow - \int \frac{1-u}{u^2} \partial u = \int \frac{t+1}{t^2} \partial t \Leftrightarrow$$

$$\Leftrightarrow - \left(\int \frac{1}{u^2} \partial u - \int \frac{u}{u^2} \partial u \right) = \int \frac{t}{t^2} \partial t + \int \frac{1}{t^2} \partial t$$

$$\Leftrightarrow - \left(\int u^{-2} \partial u - \int \frac{1}{u} \partial u \right) = \int \frac{1}{t} \partial t + \int t^{-2} \partial t$$

$$\Leftrightarrow + \frac{u^{-1}}{1} + \ln|u| = \ln|t| + \frac{t^{-1}}{-1} + C, \quad C \in \mathbb{R}$$

$$\Leftrightarrow \ln|u| - \ln|t| = \frac{1}{u} - \frac{1}{t} + C \Leftrightarrow \ln \left| \frac{u}{t} \right| = \frac{1}{u} - \frac{1}{t} + C$$

$$\Leftrightarrow \left| \frac{u}{t} \right| = e^{\frac{1}{u} - \frac{1}{t}} \cdot e^C \Leftrightarrow \left| \frac{u}{t} \right| = e^{\frac{1}{u} - \frac{1}{t}} \cdot \kappa, \quad \kappa \in \mathbb{R}^+$$

$$\Leftrightarrow \frac{u}{t} = e^{\frac{1}{u} - \frac{1}{t}} \cdot \kappa, \quad \kappa \in \mathbb{R} \setminus \{0\} \Leftrightarrow u = e^{\frac{1}{u} - \frac{1}{t}} \cdot t \cdot \kappa, \quad \kappa \in \mathbb{R} \setminus \{0\}$$

$u=0$ também é solução (singular)

logo $u = \kappa e^{\frac{1}{u} - \frac{1}{t}} \cdot t$, $\kappa \in \mathbb{R}$ é solução geral da EDO

$$d) (n^2 - 1)y' + 2ny^2 = 0 \quad (\Rightarrow) \quad (n^2 - 1)y' = -2ny^2 \quad (\Rightarrow) \quad n^2 y' - y' = -2ny^2 \quad (\Rightarrow) \quad n^2 \frac{dy}{dn} = -2ny^2$$

$$(\Rightarrow) \quad n^2 dy = -2ny^2 dn \quad (\Rightarrow) \quad dy \frac{1}{y^2} = -\frac{2n}{n^2} dn$$

$$(\Rightarrow) \int -\frac{2n}{n^2} dn = \int \frac{1}{y^2} dy \quad (\Rightarrow) \quad -2 \int \frac{1}{n} dn = \int y^{-2} dy \quad (\Rightarrow) \quad -2 \ln|n| = \frac{y^{-1}}{-1} + C \quad (\Rightarrow)$$

$$(\Rightarrow) -\frac{1}{y} = -2 \ln|n| + C \quad (\Rightarrow) \quad \frac{1}{y} = 2 \ln|n| + C \quad (\Rightarrow) \quad y = \frac{1}{2 \ln|n| + C}$$

⑧ a) $y' + 2y = \cos n$

$$1) \int 2 dn = 2n \rightarrow \mu(n) = e^{2n}$$

$$2) (e^{2n} \cdot y)' = e^{2n} \cdot \cos n \quad (\Rightarrow) \quad e^{2n} \cdot y = \int e^{2n} \cdot \cos n \, dn \quad (\Rightarrow) \dots (\Rightarrow) e^{2n} \cdot y = \frac{e^{2n}(\sin(n) + 2\cos(n))}{5}$$

$$(\Rightarrow) y = \frac{e^{2n}(\sin(n) + 2\cos(n))}{e^{2n} \cdot 5} + \frac{C}{e^{2n}} \quad (\Rightarrow) \quad y = \frac{\sin(n)}{5} + \frac{2\cos(n)}{5} + Ce^{-2n}, \quad C \in \mathbb{R}$$

b) $n^3 y' - y - 1 = 0 \quad (\Rightarrow) \quad y' - \frac{1}{n^3} y = \frac{1}{n^3}$

c) $\frac{1}{n} y' - \frac{1}{n^2 + 1} y = \frac{\sqrt{n^2 + 1}}{n}, \quad n \neq 0$

$\mu(n) \int -\frac{1}{n^3} dn = -\int n^{-3} dn = -\frac{n^{-2}}{-2} + C = \frac{1}{2n^2} + C$

$\mu(n) = e^{\frac{1}{2n^2}}$

$$\left(e^{\frac{1}{2n^2}} \cdot y\right)' = e^{\frac{1}{2n^2}} \cdot \frac{1}{n^3} \quad (\Rightarrow) \quad e^{\frac{1}{2n^2}} \cdot y = \int e^{\frac{1}{2n^2}} \cdot \frac{1}{n^3} \, dn$$

$$(\Rightarrow) e^{\frac{1}{2n^2}} \cdot y = -e^{\frac{1}{2n^2}} + C$$

$$(\Rightarrow) y = -1 + C \cdot e^{\frac{1}{2n^2}}, \quad C \in \mathbb{R}$$

$$\left(\frac{1}{2n^2}\right)' = \frac{1}{2} (n^{-2})' = \frac{-2n^{-3}}{2} = -\frac{1}{n^3}$$

$$c) \quad \frac{1}{n} y' - \frac{1}{n^2+1} y = \frac{\sqrt{n^2+1}}{n}, \quad n \neq 0$$

$$(\Rightarrow) y' - \frac{n}{n^2+1} y = \sqrt{n^2+1}$$

$\mu(n)$:

$$-\int \frac{n}{n^2+1} \, dn = -\frac{1}{2} \int \frac{2n}{n^2+1} \, dn = -\frac{1}{2} \ln|n^2+1| + C$$

$$\mu(n) = e^{-\frac{1}{2} \ln|n^2+1|} = e^{\ln(n^2+1)^{-1/2}} = e^{\ln\left(\frac{1}{\sqrt{n^2+1}}\right)} = \frac{1}{\sqrt{n^2+1}}$$

$$\left(\frac{1}{\sqrt{n^2+1}} y\right)' = \sqrt{n^2+1} \cdot \frac{1}{\sqrt{n^2+1}} \quad (\Rightarrow) \quad \frac{1}{\sqrt{n^2+1}} y = \int 1 \, dn \quad (\Rightarrow) \quad \frac{1}{\sqrt{n^2+1}} y = n + C$$

$$(\Rightarrow) y = n\sqrt{n^2+1} + C\sqrt{n^2+1}, \quad C \in \mathbb{R}$$

9) $x^2 y' + 2xy = 1$ em $]0, +\infty[$. Qualquer solução tende para zero quando $x \rightarrow +\infty$?

$$1) \quad y' + \frac{2x}{x^2} y = \frac{1}{x^2}$$

$$2) \quad \int \frac{2x}{x^2} dx = 2 \int \frac{1}{x} dx = 2 \ln|x| + C \rightarrow \mu(x) = e^{2 \ln|x|} = e^{\ln(x^2)} = x^2$$

$$3) \quad (x^2 \cdot y)' = \frac{1}{x^2} \cdot x^2 \quad (\Rightarrow) \quad x^2 \cdot y = \int 1 dx \quad (\Rightarrow) \quad x^2 \cdot y = x + C \quad (\Rightarrow) \quad y = \frac{x}{x^2} + \frac{C}{x^2} \quad (\Rightarrow) \quad y = \frac{1}{x} + \frac{C}{x^2}, C \in \mathbb{R}$$

$$\lim_{x \rightarrow +\infty} \frac{1}{x} + \frac{C}{x^2} = 0 + 0 = 0 \quad \checkmark$$

10) a) $(x^2 + y^2) y' = xy \quad (\Rightarrow) \quad y' = \frac{xy}{x^2 + y^2}$

$$f(x, y) = \frac{xy}{(x^2 + y^2)}$$

$$f(\lambda x, \lambda y) = \frac{\lambda x \lambda y}{(\lambda x)^2 + (\lambda y)^2} = \frac{\lambda^2 xy}{\lambda^2 x^2 + \lambda^2 y^2} = \frac{\cancel{\lambda^2} xy}{\cancel{\lambda^2} (x^2 + y^2)} = \frac{xy}{x^2 + y^2} = f(x, y) \quad \checkmark$$

MV: $z = \frac{y}{x} \quad (\Rightarrow) \quad y = zx$

$$y' = z'x + z \rightarrow z'x + z = \frac{x \cdot zx}{x^2 + (zx)^2} \quad (\Rightarrow) \quad z'x + z = \frac{\cancel{zx^2}}{\cancel{x^2} (1 + z^2)} \quad (\Rightarrow) \quad z'x + z = \frac{z}{1 + z^2} \quad (\Rightarrow)$$

$$(\Rightarrow) \quad \frac{\partial z}{\partial x} x = \frac{z}{1 + z^2} - z \quad (\Rightarrow) \quad \frac{\partial z}{\partial x} x = \frac{z - z(1 + z^2)}{1 + z^2} \quad (\Rightarrow)$$

$$\Leftrightarrow \frac{\partial z}{\partial u} u = \frac{z - z + z^3}{1+z^2} \Leftrightarrow \frac{\partial z}{\partial u} u = \frac{-z^3}{1+z^2} \Leftrightarrow \frac{1}{u} du = -\frac{1+z^2}{z^3} dz$$

$$\int \frac{1}{u} du = - \int \frac{1+z^2}{z^3} dz \Leftrightarrow \int \frac{1}{u} du = - \left(\int \frac{1}{z^3} dz + \int \frac{z^2}{z^3} dz \right) \Leftrightarrow \ln|u| + c = -\frac{z^{-2}}{-2} + \ln|z|$$

$$\Leftrightarrow \ln|z| + \frac{z^{-2}}{-2} = -\ln|u| + c \Leftrightarrow \ln|z| - \frac{1}{2z^2} = -\ln|u| + c, \quad c \in \mathbb{R}$$

$$\text{MV: } z = \frac{y}{u}$$

$$\ln\left|\frac{y}{u}\right| - \frac{1}{2\left(\frac{y}{u}\right)^2} = -\ln|u| + c \Leftrightarrow \ln(y) - \ln(u) - \frac{1}{2} \cdot \frac{1}{\frac{y^2}{u^2}} = -\ln|u| + c$$

$$\Leftrightarrow \ln(y) - \frac{u^2}{2y^2} = -\ln|u| - \ln|u| + c \Leftrightarrow \ln(y) = -2\ln(u) + \frac{u^2}{2y^2} \Leftrightarrow$$

$$\Leftrightarrow y = e^{-2\ln|u| + \frac{u^2}{2y^2} + c} \Leftrightarrow y = e^{\ln|u|^2} \cdot e^{\frac{u^2}{2y^2}} \cdot e^c \Leftrightarrow$$

$$\Leftrightarrow y = u^2 \cdot e^{\frac{u^2}{2y^2}} \cdot c, \quad c \in \mathbb{R} \setminus \{0\}$$

$$b) y' \left(1 - \ln\left(\frac{y}{u}\right)\right) = \frac{y}{u}, \quad u > 0$$

$$\Leftrightarrow y' = \frac{\frac{y}{u}}{1 - \ln\left(\frac{y}{u}\right)} \Leftrightarrow y' = \frac{y}{u - u \ln\left(\frac{y}{u}\right)}$$

1) homogenea?

$$f(x, y) = \frac{y}{x - x \ln\left(\frac{y}{x}\right)}$$

$$f(\lambda x, \lambda y) = \frac{\lambda y}{\lambda x - \lambda x \ln\left(\frac{\lambda y}{\lambda x}\right)} = \frac{\lambda y}{\lambda x (1 - \ln\left(\frac{y}{x}\right))} = \frac{y}{x (1 - \ln\left(\frac{y}{x}\right))} = f(x, y) \checkmark$$

2) MV: $z = \frac{y}{x} \Leftrightarrow y = zx \quad y' = z'x + z$

$$z'x + z = \frac{\frac{zx}{x}}{1 - \ln\left(\frac{zx}{x}\right)} \quad (\Rightarrow) \quad z'x + z = \frac{z}{1 - \ln(z)} \quad (\Rightarrow) \quad z'x = \frac{z}{1 - \ln(z)} - \frac{z(1 - \ln(z))}{1 - \ln(z)}$$

$$(\Rightarrow) z'x = \frac{z - z + z \ln(z)}{1 - \ln(z)} \quad (\Rightarrow) \quad z'x = \frac{z \ln(z)}{1 - \ln(z)} \quad (\Rightarrow) \quad \frac{1}{x} dx = \frac{1 - \ln(z)}{z \ln(z)} dz$$

$$3) \int \frac{1}{x} dx = \int \frac{1 - \ln(z)}{z \ln(z)} dz \quad (\Rightarrow) \quad \ln|x| + C = \int \frac{1}{z \ln(z)} dz - \int \frac{\ln(z)}{z \ln(z)} dz$$

$$(\Rightarrow) \ln|x| + C = \ln|\ln|z|| - \ln(z)$$

MV: $z = \frac{y}{x}$

$$\ln\left(\ln\left|\frac{y}{x}\right|\right) - \ln\left(\frac{y}{x}\right) = \ln|x| + C \quad (\Rightarrow) \quad \ln(\ln(y) - \ln(x)) - \ln(y) - \ln(x) = \ln|x| + C$$

$$(\Rightarrow) \ln(\ln(y) - \ln(x)) = 2 \ln|x| + \ln(y) + C \quad (\Rightarrow) \quad \ln\left(\frac{y}{x}\right) = e^{2 \ln|x|} \cdot e^{\ln|y|} \cdot e^C \quad (\Rightarrow)$$

$$\Leftrightarrow \ln(y) - \ln(n) = n^2 \cdot y \cdot \kappa, \quad \kappa \in \mathbb{R}^+ \Leftrightarrow \ln(y) = n^2 \cdot y \cdot \kappa + \ln(n) \Leftrightarrow$$

$$\Leftrightarrow y = e^{n^2 y \kappa} \cdot e^{\ln(n)} \Leftrightarrow y = e^{\kappa y n^2} \cdot n, \quad \kappa \in \mathbb{R}^+$$



$$(11) \quad y' = \frac{y}{n} (1 + \ln y - \ln n), \quad n, y \in \mathbb{R}^+$$

$$a) \quad f(n, y) = \frac{y}{n} (1 + \ln y - \ln n)$$

$$f(\lambda n, \lambda y) = \frac{\lambda y}{\lambda n} (1 + \ln(\lambda y) - \ln(\lambda n)) = \frac{y}{n} \left(1 + \ln\left(\frac{\lambda y}{\lambda n}\right)\right) = \frac{y}{n} \left(1 + \ln\left(\frac{y}{n}\right)\right) = \frac{y}{n} (1 + \ln(y) - \ln(n))$$

MV

$$z = \frac{y}{n} \Leftrightarrow y = zn$$

$$y' = z'n + z$$

$$z'n + z = \frac{zn}{n} (1 + \ln(zn) - \ln(n)) \Leftrightarrow z'n + z = z (1 + \ln(z) + \ln(n) - \ln(n)) \Leftrightarrow z'n + z = z (1 + \ln(z)) \Leftrightarrow$$

$$\Leftrightarrow \frac{\partial z}{\partial n} n = z + z \ln(z) - z \Leftrightarrow \frac{\partial z}{\partial n} n = z \ln(z)$$

Resolven variaveis sep.

$$\frac{1}{z \ln(z)} dz = \frac{1}{u} du \quad (\Rightarrow) \quad \int \frac{1}{z \ln(z)} dz = \int \frac{1}{u} du \quad (\Rightarrow) \quad \int \frac{\frac{1}{z}}{\ln(z)} dz = \ln|u| + C \quad (\Rightarrow)$$

$$(\Rightarrow) \ln|\ln(z)| = \ln|u| + C, \quad C \in \mathbb{R}$$

$$(\Rightarrow) \ln|z| = e^{\ln|u| + C} \quad (\Rightarrow) \ln|z| = C|u| \quad (\Rightarrow) |z| = e^{C|u|}, \quad C \in \mathbb{R}^+ \quad (\Rightarrow) z = e^{C u}, \quad C \in \mathbb{R} \setminus \{0\}$$

$$MV: z = \frac{y}{u}$$

$$\frac{y}{u} = e^{C u} \quad (\Rightarrow) y = \frac{e^{C u}}{u}$$

12) a) $u y' + y = y^2 \ln(u) \quad (\Rightarrow) \quad y' + \frac{y}{u} = y^2 \frac{\ln(u)}{u}$

1) dividir por y^2

$$y' \cdot y^{-2} + y \cdot y^{-2} \cdot \frac{1}{u} = \frac{\ln(u)}{u} \quad (\Rightarrow) \quad y' \cdot y^{-2} + y^{-1} \cdot \frac{1}{u} = \frac{\ln(u)}{u}$$

$$2) MV: z = y^{1-\alpha} \quad (\Rightarrow) \quad z = y^{-1} \quad \rightarrow \quad z' = -1 y^{-2} y' \quad (\Rightarrow) \quad z' = -y^{-2} \cdot y'$$

$$y' \cdot y^{-2} + y^{-1} \cdot \frac{1}{u} = \frac{\ln(u)}{u} \quad \rightarrow \quad -z' + z \cdot \frac{1}{u} = \frac{\ln(u)}{u} \quad (\Rightarrow) \quad z' - z \cdot \frac{1}{u} = -\frac{\ln(u)}{u} \quad \leftarrow \text{linear } 1^{\text{a}} \text{ ordem}$$

$$3) \mu(x) = e^{\int p(x) dx}$$

$$- \int \frac{1}{x} dx = -\ln|x| \rightarrow \mu(x) = e^{-\ln|x|} = x^{-1}$$

4) Mult. por $\mu(x)$

$$(x^{-1} z)' = \frac{\ln(x)}{x} \cdot x^{-1} \Rightarrow \frac{z}{x} = - \int \frac{\ln(x)}{x^2} dx \Rightarrow \frac{z}{x} = - \left(\frac{\ln(x)}{x} - \int \frac{1}{x} \cdot \frac{1}{x} dx \right) \Rightarrow$$

$$u = \ln(x) \quad dv = \frac{1}{x^2}$$

$$du = \frac{1}{x}$$

$$v = \frac{x^{-1}}{-1} \Rightarrow v = -\frac{1}{x}$$

$$\Rightarrow \frac{z}{x} = - \left(\frac{\ln(x)}{x} - \frac{x^{-1}}{-1} + C \right) \Rightarrow \frac{z}{x} = - \frac{\ln(x)}{x} + \frac{1}{x} - C$$

$$\Rightarrow z = -\ln(x) + 1 - Cx, \quad C \in \mathbb{R}$$

$$5) z = y^{-1}$$

$$y^{-1} = -\ln(x) + 1 - Cx \Rightarrow y = \frac{1}{-\ln(x) + 1 - Cx}, \quad C \in \mathbb{R}$$

$y=0$ também é solução singular

$$b) y' - \frac{y}{2x} = 5x^2 y^5, \quad x \neq 0$$

1) Dividir por y^5

$$y' \cdot y^{-5} - y^{-4} \cdot \frac{1}{2x} = 5x^2.$$

2) MV: $z = y^{1-5} \Leftrightarrow z = y^{-4} \quad z' = -4y^{-5} y' \Leftrightarrow \frac{z'}{-4} = y^5 y'$

$$\frac{z'}{-4} - z \cdot \frac{1}{2x} = 5x^2 \quad \Leftrightarrow \quad z' + z \frac{4}{2x} = 20x^2$$

3) $\int \frac{4}{2x} dx = 2 \int \frac{1}{x} dx = 2 \ln|x| + C \rightarrow \mu(x) = e^{2 \ln|x|} = e^{\ln(x)^2} = x^2$

$$(x^2 \cdot z)' = 20x^2 \cdot x^2 \quad \Leftrightarrow \quad x^2 \cdot z = \int 20x^4 dx \quad \Leftrightarrow \quad x^2 \cdot z = 20 \cdot \frac{x^5}{5} + C$$

$$\Leftrightarrow x^2 \cdot z = 4x^5 + C \quad \Leftrightarrow \quad z = 4x^5 \cdot x^{-2} + \frac{C}{x^2} \quad \Leftrightarrow \quad z = 4x^3 + \frac{C}{x^2}, \quad C \in \mathbb{R}$$

4) $z = y^{-4}$

$$y^{-4} = 4x^3 + \frac{C}{x^2} \quad \Leftrightarrow \quad y^{-4} = \frac{4x^5 + C}{x^2} \quad \Leftrightarrow \quad y^4 = \frac{x^2}{4x^5 + C}, \quad C \in \mathbb{R}$$

$y = 0$ também é solução (singular)

Revisões Cap 3 - Ficha 3

9) Transformada de Laplace de f ?

a) $f(t) = 2\sin(3t) + t - 5e^{-t}$

$$\begin{aligned}\mathcal{L}\{f(t)\}(s) &= 2\mathcal{L}\{\sin(3t)\}(s) + \mathcal{L}\{t\} - 5\mathcal{L}\{e^{-t}\}(s) \\ &= 2\frac{3}{s^2 + 3^2} + \frac{1}{s^2} - 5\frac{1}{s+1}, \quad s > -1, s > 0, s > 0 \\ &= \frac{6}{s^2 + 9} + \frac{1}{s^2} - \frac{5}{s+1}, \quad s > 0\end{aligned}$$

b) $f(t) = e^{-t} \sinh(5t)$

$$\mathcal{L}\{f(t)\}(s) = \mathcal{L}\{\sinh(5t)\}(s+1) = \frac{5}{(s+1)^2 - 25}, \quad s > 4$$

c) $f(t) = e^{2t} \cos(5t)$

$$\mathcal{L}\{f(t)\}(s) = \mathcal{L}\{\cos(5t)\}(s-2) = \frac{5}{(s-2)^2 + 5^2}, \quad s > 2$$

d) $f(t) = e^{3t}$

$$\mathcal{L}\{f(t)\}(s) = \frac{1}{s-3}, \quad s > 3$$

e) $f(t) = t \cosh(t)$

$$\mathcal{L}\{f(t)\}(s) = (-1)' \mathcal{L}\{\cosh(t)\}'(s) = -1 \left(\frac{s}{s^2 - 1} \right)' = -1 \left(\frac{(s^2 - 1) - 2s \cdot s}{(s^2 - 1)^2} \right) = -\frac{s^2 - 2s^2 - 1}{(s^2 - 1)^2} = \frac{s^2 + 1}{(s^2 - 1)^2}, \quad s > 1$$

$$f) f(t) = \pi - 5e^{-t} t^{10}$$

$$\mathcal{L}\{f(t)\} = \mathcal{L}\{\pi\} - 5 \mathcal{L}\{t^{10}\}(s+1) = \frac{1}{s} - 5 \frac{10!}{(s+1)^{11}}, s > 0$$

$$g) f(t) = (3t-1)\sin t \Rightarrow 3t\sin t - \sin t$$

$$\begin{aligned} \mathcal{L}\{f(t)\} &= 3\mathcal{L}\{t\sin t\} - \mathcal{L}\{\sin t\} = 3\left(-1 \mathcal{L}\{\sin t\}'(s)\right) - \frac{1}{s^2+1}, s > 0 \\ &= 3\left(-\left(\frac{1}{s^2+1}\right)'\right) - \frac{1}{s^2+1} = \\ &= 3\left(\frac{-2s}{(s^2+1)^2}\right) - \frac{1}{s^2+1} = \frac{6s}{(s^2+1)^2} - \frac{1}{s^2+1}, s > 0 \end{aligned}$$

$$h) f(t) = (1 - H_{\pi}(t))\sin t = \sin t - H_{\pi}(t)\sin t = \sin t + H_{\pi}(t)(\sin t - \pi)$$

$$= \mathcal{L}\{\sin t\} - \mathcal{L}\{\sin t\} \cdot e^{-\pi s} = \frac{1}{s^2+1} - \frac{1}{s^2+1} \cdot e^{-\pi s}, s > 0$$

$$(10) a) f(s) = \frac{2s}{s^2-9}$$

$$2\mathcal{L}^{-1}\left\{\frac{s}{s^2-3^2}\right\} = 2\left(\frac{s}{s^2-3^2}\right) = 2\cosh(3t)$$

$$b) f(s) = \frac{4}{s^7}$$

$$\mathcal{L}^{-1}\{f(s)\} = 4 \mathcal{L}^{-1}\left\{\frac{1}{s^7}\right\} = 4 \frac{1}{6!} \mathcal{L}^{-1}\left\{\frac{6!}{s^7}\right\} = \frac{4}{6!} t^6$$

$$c) f(s) = \frac{1}{s^2+6s+9} = \frac{1}{(s+3)^2}$$

$$\mathcal{L}^{-1}\left\{\frac{1}{(s+3)^2}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} \cdot e^{-3t} = t \cdot e^{-3t}$$

$$d) \frac{1}{s^2+s-2} = \frac{1}{(s-1)(s+2)} = \frac{A}{s-1} + \frac{B}{s+2} = \frac{A(s+2) + B(s-1)}{(s-1)(s+2)}$$

$$\begin{cases} A+B=0 \\ 2A-B=1 \end{cases} \quad \begin{cases} A=-B \\ 3A=1 \end{cases} \quad \begin{cases} B=-\frac{1}{3} \\ A=\frac{1}{3} \end{cases}$$

$$\mathcal{L}^{-1}\left\{\frac{\frac{1}{3}}{s-1}\right\} - \mathcal{L}^{-1}\left\{\frac{\frac{1}{3}}{s+2}\right\} = \frac{1}{3} \mathcal{L}^{-1}\left\{\frac{1}{s-1}\right\} - \frac{1}{3} \mathcal{L}^{-1}\left\{\frac{1}{s+2}\right\} = \frac{1}{3} e^t - \frac{1}{3} e^{-2}$$

$$e) \frac{1}{s^2+4s+6} = \frac{1}{s^2+4s+4+2} = \frac{1}{(s+2)^2+2}$$

$$\mathcal{L}^{-1}\left\{\frac{1}{(s+2)^2+2}\right\} = e^{-2t} \mathcal{L}^{-1}\left\{\frac{1}{s^2+2}\right\} = e^{-2t} \mathcal{L}^{-1}\left\{\frac{1}{s^2+\sqrt{2}^2}\right\} = e^{-2t} \sin(\sqrt{2}t)$$

$$f) \frac{3s-1}{s^2-4s+13} = \frac{3s-1}{s-4s+4+9} = \frac{3s-1}{(s-2)^2+3^2}$$

$$\mathcal{L}^{-1}\left\{\frac{3s-1}{(s-2)^2+3^2}\right\} = \mathcal{L}^{-1}\left\{\frac{3s}{(s-2)^2+3^2}\right\} - \mathcal{L}^{-1}\left\{\frac{1}{(s-2)^2+3^2}\right\} = e^{2t} \mathcal{L}^{-1}\left\{\frac{s-2}{s^2+3^2}\right\} - e^{2t} \mathcal{L}^{-1}\left\{\frac{1}{s^2+3^2}\right\}$$

$$= e^{2t} \cdot 3 \mathcal{L}^{-1}\left\{\frac{s}{s^2+3^2}\right\} - e^{2t} \cdot 3 \mathcal{L}^{-1}\left\{\frac{1}{s^2+3^2}\right\} = e^{2t} \cos(3t) - e^{2t} \sin(3t)$$

$$= e^{2t} \cdot 3 \cos(3t) - e^{2t} \cdot \frac{1}{2} \sin(3t) - e^{2t} \sin(3t)$$

$$= e^{2t} \left(3 \cos(3t) - \frac{\sin(3t)}{2} - \sin(3t) \right) = e^{2t} \left(3 \cos(3t) - \frac{3 \sin(3t)}{2} \right)$$

$$(11) \quad a) \int_0^{+\infty} t^{10} e^{-2t} dt = \mathcal{L}\{t^{10}\}(s=2) = \frac{10!}{2^{11}}$$

$$b) \int_0^{+\infty} e^{-3t} t \sin(t) dt = \mathcal{L}\{t \sin(t)\}(3) = (-1) \mathcal{L}\{\sin(t)\}(3) = -1 \left(\frac{1}{s^2+1} \right)' \Big|_{s=3} = \frac{-2s}{(s^2+1)^2} \Big|_{s=3} = \frac{6}{10^2} = \frac{6}{100} = \frac{3}{50}$$

$$(12) \quad f'(t) + 2f(t) = e^t \quad f(0)=2 \quad f(t)=?$$

$$\mathcal{L}\{f'(t) + 2f(t)\} = \mathcal{L}\{e^t\} \quad (\Rightarrow) \quad \mathcal{L}\{f'(t)\} + 2\mathcal{L}\{f(t)\} = \frac{1}{s-1}$$

$$(\Rightarrow) s \mathcal{L}\{f(t)\} - s^0 f(0) + 2\mathcal{L}\{f(t)\} = \frac{1}{s-1}$$

$$(\Rightarrow) s \mathcal{L}\{f(t)\} - 2 + 2\mathcal{L}\{f(t)\} = \frac{1}{s-1} \quad (\Rightarrow) \quad s \mathcal{L}\{f(t)\} + 2\mathcal{L}\{f(t)\} = \frac{1}{s-1} + 2$$

$$(\Rightarrow) \mathcal{L}\{f(t)\}(s+2) = \frac{1}{s-1} + 2 \quad (\Rightarrow) \quad \mathcal{L}\{f(t)\} = \frac{1+2s-2}{s-1} \cdot \frac{1}{s+2}$$

$$(\Rightarrow) \mathcal{L}\{f(t)\} = \frac{2s-1}{(s-1)(s+2)} \quad (\Rightarrow) \quad f(t) = \mathcal{L}^{-1}\left\{ \frac{2s-1}{(s-1)(s+2)} \right\}$$

$$\mathcal{L}^{-1}\left\{ \frac{2s-1}{(s-1)(s+2)} \right\} =$$

$$\frac{2s-1}{(s-1)(s+2)} = \frac{A}{s-1} + \frac{B}{s+2} = \frac{A(s+2) + B(s-1)}{(s-1)(s+2)}$$

$$\begin{cases} A+B=2 \\ 2A-B=-1 \end{cases}$$

$$\begin{cases} A=2-B \\ 4-2B-B=-1 \end{cases}$$

$$\begin{cases} -3B=-5 \\ B=\frac{5}{3} \end{cases}$$

$$\begin{cases} A=\frac{1}{3} \\ B=\frac{5}{3} \end{cases}$$

$$\mathcal{L}^{-1}\left\{\frac{\frac{1}{3}}{s-1}\right\} + \mathcal{L}^{-1}\left\{\frac{\frac{5}{3}}{s+2}\right\} = \frac{1}{3}\mathcal{L}^{-1}\left\{\frac{1}{s-1}\right\} + \frac{5}{3}\mathcal{L}^{-1}\left\{\frac{1}{s+2}\right\} = \frac{1}{3}e^t + \frac{5}{3}e^{-2t}t, \quad t > 0$$

$$(2) \int_1^{+\infty} ne^{-n} dn = ?$$

$$\lim_{t \rightarrow +\infty} \int_1^t ne^{-n} dn = \lim_{t \rightarrow +\infty} -ne^{-n} \Big|_1^t - \int_1^t -e^{-n} dn = \lim_{t \rightarrow +\infty} -ne^{-n} \Big|_1^t - \left[e^{-n} \right]_1^t$$

$$u=n \quad dv=e^{-n}$$

$$du=1 \quad v=-e^{-n}$$

$$= \lim_{t \rightarrow +\infty} -te^{-t} - (-1e^{-1}) - (e^{-t} - e^{-1}) = \lim_{t \rightarrow +\infty} -te^{-t} + e^{-1} - e^{-t} + e^{-1} =$$

$$= \frac{2}{e} \quad \text{conv.}$$

$$\alpha > 1 \quad \text{conv}$$

$$\beta < 0 \quad \text{conv}$$

$$(6) \quad a) \int_1^{+\infty} \frac{\sin^2 n}{n^{5/2}}$$

$$\frac{1}{n^{5/2}} \quad \left(\frac{1}{n^\alpha}, \alpha > 1 \text{ conv} \right)$$

$$\frac{\sin^2 n}{n^{5/2}} < \frac{1}{n^{5/2}} \rightarrow \text{pelo C. Comparação é conv}$$

$$b) \int_1^{+\infty} \frac{2n}{e^{2n}-1}$$

$$b(n) = \frac{1}{e^{2n}} \quad (e^{\beta n} \quad (\beta < 0) \text{ conv})$$

$$L = \lim_{n \rightarrow +\infty} \frac{\frac{2n}{e^{2n}-1}}{\frac{1}{e^{2n}}} = \lim_{n \rightarrow +\infty} \frac{2ne^{2n}}{e^{2n}-1} = \lim_{n \rightarrow +\infty} \frac{\cancel{4e^{2n}}}{\cancel{2e^{2n}}} = \lim_{n \rightarrow +\infty} 2$$

Pelo C. Limite como $L \in \mathbb{R}^+$
e $g(n)$ conv ent conv.

$$c) \int_1^{+\infty} \frac{5n^2-3}{n^8+n-1} \quad g(n) = \frac{n^2}{n^8} = \frac{1}{n^6} \quad (\text{conv})$$

$$L = \lim \frac{\frac{5n^2-3}{n^8+n-1}}{\frac{1}{n^6}} = \lim \frac{n^6(5n^2-3)}{n^8+n-1} = \lim \frac{5n^8-3n^6}{n^8+n-1} = 5 \quad \text{conv.}$$

$$d) \int_0^{+\infty} e^{n^2} dn > e^n \quad (p=0 \text{ div}) \text{ etwa div}$$

$$13) \int_1^{+\infty} \frac{1}{(1+n^2) \arctg n} dn = \int_1^{+\infty} \frac{1}{u} du = \left[\ln |\arctg n| \right]_1^{+\infty} = \lim_{t \rightarrow +\infty} \left[\ln |\arctg n| \right]_1^t = \lim_{t \rightarrow +\infty} \left(\ln(\arctg(t)) - \ln(\arctg(1)) \right)$$

$$= \lim_{t \rightarrow +\infty} \ln \left(\frac{\arctg(t)}{\arctg(1)} \right) = \lim_{t \rightarrow +\infty} \ln \left(\frac{\frac{\pi}{2}}{\frac{\pi}{4}} \right) = \ln \left(\frac{4\pi}{2\pi} \right) = \ln(2) \quad \text{conv}$$

$$14) \int_{-\infty}^0 e^n (4-n) dn = - \int_0^{+\infty} e^{-u} (4+u) (-du) = \int_0^{+\infty} e^{-u} (4+u) du =$$

$u = -n$
 $dn = -du$

n	u
$-\infty$	$+\infty$
0	0

$$15) f(n) = \frac{1}{n \sqrt{\ln n}}$$

$$\int \frac{1}{n \sqrt{\ln n}} dn = \int \frac{1}{n} \cdot \ln(n)^{-1/2} dn = \frac{\ln(n)^{1/2}}{\frac{1}{2}} = 2 \sqrt{\ln(n)} + C$$

$$f(e) \rightarrow 2 \sqrt{\ln(e)} + C = 0 \quad (\Rightarrow) \quad 2 \sqrt{\ln(e)} = -C \quad (\Rightarrow) \quad \sqrt{\ln(e)} = -\frac{C}{2} \quad (\Rightarrow) \quad \sqrt{1} = -\frac{C}{2} \quad (\Rightarrow) \quad 1 = -\frac{C}{2} \quad (\Rightarrow) \quad 2 = -C \quad (\Rightarrow) \quad C = -2$$

$$\therefore 2\sqrt{\ln(n)} - 2$$

(19) a) $3u' - u = \cos t$, $u(0) = -1$

$$\mathcal{L}\{3u'\} - \mathcal{L}\{u\} = \mathcal{L}\{\cos t\} \quad (\Rightarrow) \quad 3\mathcal{L}\{u'\} - \mathcal{L}\{u\} = \frac{s}{s^2+1}$$

$$(\Rightarrow) 3(s\mathcal{L}\{u\} - s^0 \cdot -1) - \mathcal{L}\{u\} = \frac{s}{s^2+1} \quad (\Rightarrow) \quad 3s\mathcal{L}\{u\} + 3 - \mathcal{L}\{u\} = \frac{s}{s^2+1}$$

$$(\Rightarrow) \mathcal{L}\{u\}(3s-1) = \frac{s}{s^2+1} - \frac{3(s^2+1)}{s^2+1} \quad (\Rightarrow) \quad \mathcal{L}\{u\} = \frac{s-3s^2-3}{s^2+1} \cdot \frac{1}{3s-1} \quad (\Rightarrow) \quad \mathcal{L}\{u\} = \frac{-3s^2+s-3}{(s^2+1)(3s-1)}$$

$$(\Rightarrow) u = \mathcal{L}^{-1}\left\{\frac{-3s^2+s-3}{(s^2+1)(3s-1)}\right\}$$

$$\mathcal{L}^{-1}\left\{\frac{-3s^2+s-3}{(s^2+1)(3s-1)}\right\} =$$

C.A

$$\frac{-3s^2+s-3}{(s^2+1)(3s-1)} = \frac{A}{3s-1} + \frac{Bs+C}{s^2+1} = \frac{A(s^2+1) + B(3s^2-s) + C(3s-1)}{(3s-1)(s^2+1)}$$

$$\begin{cases} A+3B = -3 \\ -B+3C = 1 \\ A-C = -3 \end{cases}$$

$$\begin{cases} B = \frac{-3-A}{3} \\ 3C = 1+B \\ A = -3+C \end{cases}$$

$$\begin{cases} 3C = 1 + \frac{-3-A}{3} \end{cases}$$

$$\begin{cases} C = \frac{1}{3} + \frac{-3-A}{9} \\ A = -3 + \frac{1}{3} - \frac{3-A}{9} \end{cases}$$

$$\begin{cases} A = -\frac{8}{3} - \frac{1}{3} - \frac{A}{9} \\ A = -\frac{9}{3} - \frac{A}{9} \end{cases}$$

$$\begin{cases} A = -3 - \frac{A}{9} \end{cases}$$

$$\begin{cases} \frac{9A}{9} + \frac{A}{9} = -3 \end{cases}$$

$$\begin{cases} \frac{10A}{9} = -3 \end{cases}$$

$$\begin{cases} A = -\frac{3 \cdot 9}{10} \end{cases}$$

$$\begin{cases} B = -\frac{3 \cdot 10 + \frac{27}{10}}{3} \\ A = -\frac{27}{10} \end{cases}$$

$$\begin{cases} B = \frac{-3}{3} \\ 3C = 1 - \frac{1}{10} \end{cases}$$

$$\begin{cases} B = -\frac{1}{10} \\ 3C = \frac{10}{10} - \frac{1}{10} \\ A = -\frac{27}{10} \end{cases} \quad \begin{cases} B = -\frac{1}{10} \\ C = \frac{9}{30} \end{cases} \quad \begin{cases} B = -\frac{1}{10} \\ C = \frac{9}{30} \\ A = -\frac{27}{10} \end{cases} \quad X$$

$$b) \frac{d^2 y}{dt^2} + 36y = 0, \quad y(0) = -1 \quad \frac{dy}{dt}(0) = 2$$

$$\mathcal{L} \left\{ \frac{d^2 y}{dt^2} \right\} + 36 \mathcal{L} \{ y \} = \mathcal{L} \{ 0 \}$$

$$s^2 \mathcal{L} \{ y \} - s \cdot (-1) - s^0 \cdot 2 + 36 \mathcal{L} \{ y \} = 0$$

$$\Rightarrow s^2 \mathcal{L} \{ y \} + s - 2 + 36 \mathcal{L} \{ y \} = 0 \Rightarrow \mathcal{L} \{ y \} (s^2 + 36) = 2 - s \Rightarrow \mathcal{L} \{ y \} = \frac{2-s}{s^2+36}$$

$$\Rightarrow y = \mathcal{L}^{-1} \left\{ \frac{2-s}{s^2+36} \right\}$$

$$\mathcal{L}^{-1} \left\{ \frac{2-s}{s^2+36} \right\} = \mathcal{L}^{-1} \left\{ \frac{2}{s^2+36} \right\} - \mathcal{L}^{-1} \left\{ \frac{s}{s^2+6^2} \right\} = 2 \mathcal{L}^{-1} \left\{ \frac{1}{s^2+6^2} \right\} - \cos(6t) = \frac{2}{6} \mathcal{L}^{-1} \left\{ \frac{6}{s^2+6^2} \right\} - \cos(6t)$$

$$= \frac{2}{6} \sin(6t) - \cos(6t)$$

$$y(t) = \frac{1}{3} \sin(6t) - \cos(6t)$$

Aula revisões

$$y' - y = e^{-x} y^2$$

$$\left(\frac{1}{y}\right)' =$$

1) Div por y^2

$$y' \cdot y^{-2} - y \cdot y^{-3} = e^{-x} \quad (\Rightarrow) \quad y' \cdot y^{-2} - y^{-1} = e^{-x}$$

2) MV: $z = y^{1-2} \quad (\Rightarrow) \quad z = y^{-1}, \quad z' = -y' y^{-2}$

$$-z' - z = e^{-x} \quad (\Rightarrow) \quad z' + z = -e^{-x}$$

3) $\int 1 \, dx = x \rightarrow \mu(x) = -e^x$

$$(z \cdot e^x)' = e^{-x} \cdot e^x \quad (\Rightarrow) \quad z \cdot e^x = -\int 1 \, dx \quad (\Rightarrow) \quad z \cdot e^x = -x + C$$

4) MV: $z = y^{-1}$

$$y^{-1} \cdot e^x = x + C \quad (\Rightarrow) \quad y^{-1} = \frac{-x + C}{e^x} \quad (\Rightarrow) \quad y = \frac{e^x}{C - x}, \quad C \in \mathbb{R}$$

$$y'' + 4y = \sin x$$

$$y = y_h + y_p$$

$$y_h) \quad n^2 + 4 = 0 \quad (\Rightarrow) \quad n^2 = -4 \quad (\Rightarrow) \quad n = \pm\sqrt{-4} \quad (\Rightarrow) \quad n = \pm 2i \quad \begin{matrix} \alpha=0 \\ \beta=2 \end{matrix}$$

$$\left\{ e^{0n} \cos(2n), e^{0n} \sin(2n) \right\} \rightarrow C_1 \cos(2n) + C_2 \sin(2n), C_1, C_2 \in \mathbb{R}$$

$$y_p) \quad b(n) = \sin(n) \quad 0 + 1i \quad X$$

$$n^0 e^{\alpha n} (Q(n) \cos(\beta n) + R(n) \sin(\beta n))$$

$$y = Q \cos(n) + R \sin(n)$$

$$y' = -Q \sin(n) + R \cos(n)$$

$$y'' = -Q \cos(n) - R \sin(n)$$

$$y'' + 4y = \sin n \quad (\Rightarrow) \quad -Q \cos(n) - R \sin(n) + 4(Q \cos(n) + R \sin(n)) = \sin n$$

$$(\Rightarrow) -Q \cos(n) - R \sin(n) + 4Q \cos(n) + 4R \sin(n) = \sin n$$

$$(\Rightarrow) 3Q \cos(n) + 3R \sin(n) = \sin n$$

$$\begin{cases} 3Q = 0 \\ 3R = 1 \end{cases} \quad \begin{cases} Q = 0 \\ R = \frac{1}{3} \end{cases}$$

$$y_p = \frac{1}{3} \sin(n)$$

$$y = y_h + y_p \rightarrow y = C_1 \cos(2n) + C_2 \sin(2n) + \frac{1}{3} \sin(n), C_1, C_2 \in \mathbb{R}$$